

GPU-Accelerated Bayesian Inference of Multi-Species Biofilm Interaction Parameters via TMCMC

Full Discovery of the 15-Dimensional Interaction Matrix with Two-Phase GPU-TMCMC

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Outline

Motivation & Background

5-Species Interaction Network

Mathematical Framework

Bayesian Inference via TMCMC

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Results

Multiscale FEM Extension

Conclusions

Peri-Implantitis and Multi-Species Biofilm Dysbiosis

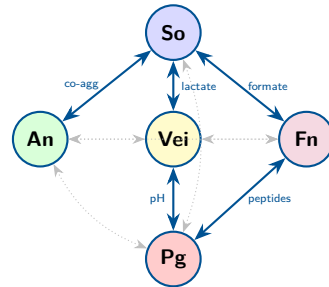
Clinical problem

- ▶ **Peri-implantitis**: inflammatory bone destruction around dental implants
- ▶ Primary driver: multi-species oral biofilm **dysbiosis** [1]
- ▶ Prevalence ~20% of implant patients [2]
- ▶ Structural mechanics of the biofilm change with community state

Open question

- ▶ **Which** inter-species interactions drive the commensal→dysbiotic transition?
- ▶ **How stiff** does the biofilm become, and what stresses does it exert?
- ▶ Current gap: no quantitative, data-driven parameter estimation framework

This work: Bayesian inference of **15 interaction parameters** from in-vitro time-series data, validated against 4 conditions



↔ Confirmed

pathway ... Free parameter

Indexing: So=1, An=2, Vei=3, Fn=4, Pg=5

Experimental design [3]

- ▶ 5 oral species: So, An, Vei, Fn, Pg
- ▶ **4 conditions:** Commensal $\times$ {Static, HOBIC}, Dysbiotic $\times$ {Static, HOBIC}
- ▶ **6 time points:** Days 1, 3, 6, 10, 15, 21
- ▶ $N = 8$ biological replicates per condition
- ▶ HOBIC = biofilm reactor mimicking gingival fluid flow

Key distinction: Pg strain

- ▶ **Commensal:** Pg ATCC 20709 (avirulent, non-encapsulated)
- ▶ **Dysbiotic:** Pg W83 (virulent, complete gingipain & fimbrial genes)

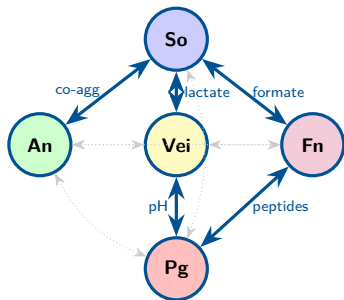
Data normalised to volume fractions $\phi_i \in [0, 1]$; Day 1 = initial condition, $N_t = 5$ usable time points per species

Species Volume Heatmap [$\times 10^6 \mu\text{m}^3$]



Heatmap: mean $\langle \phi_i(t_k) \rangle_{N=8}$ across conditions. Dysbiotic conditions show late Pg surge (Day 15–21).

Five Key Oral Species and Their Confirmed Interactions



Pair	Mechanism	Ref.
So–An	Co-aggregation; So provides growth substrates for An	[4]
So–Vei	Lactate production by So; Vei consumes lactate to elevate pH	[4]
So–Fn	Formate symbiosis; Fn bridges early–late colonisers	[4]
Vei→Pg	Vei creates alkaline micro-niche favourable to Pg (pH ↑)	[3]
Fn–Pg	Fn provides peptide nutrients; physical co-aggregation with Pg	[4]

Continuum Biofilm Model: State Variables

State variables (per material point)

- ▶ $\phi_i \in [0, 1]$: volume fraction of species i
- ▶ $\psi_i \in [0, 1]$: viability fraction (membrane-intact cells)
- ▶ $\bar{\phi}_i := \phi_i \psi_i$: *living* volume fraction
- ▶ ϕ_0 : void / fluid phase
- ▶ $\gamma(t)$: Lagrange multiplier

Volume constraint:

$$\sum_{l=0}^5 \phi_l(t) = 1 \quad \forall t$$

Key modelling choice

$\bar{\phi}_i = \phi_i \psi_i$: growth is driven by *living* cells, not total biomass

⇒ captures cell death independently of volume change

Extended Hamilton Principle [6]

$$\mathcal{H} = \int_{\tau} \left(\underbrace{\mathcal{G}}_{\text{free energy}} + \underbrace{\mathcal{C}}_{\text{constraint}} + \underbrace{\mathcal{D}}_{\text{dissipation}} \right) dt \rightarrow \text{stat}_{\xi, \gamma}$$

Free energy (nutrient-driven growth, $\alpha^* = 0$):

$$\Psi = -\frac{1}{2} c^* \bar{\phi}^\top \mathbf{A} \bar{\phi}$$

Dissipation potential:

$$\mathcal{D} = \frac{1}{2} \dot{\bar{\phi}}^\top \boldsymbol{\eta} \dot{\bar{\phi}} + \frac{1}{2} \dot{\phi}^\top \boldsymbol{\eta} \dot{\phi}$$

$$\boldsymbol{\eta} = \text{diag}(\eta_i), \quad \eta_i = 1 \quad (\text{normalised})$$

Barrier $\Psi_{\text{bar}} = K_p / [\phi_i^2 (1 - \phi_i)^2]$ enforces $\phi_i, \psi_i \in (0, 1)$

numerically.

Symmetric Interaction Matrix: 15 Free Parameters

Symmetric interaction matrix $\mathbf{A}$

$$\mathbf{A} \in \mathbb{R}^{5 \times 5}, \quad A_{ij} = A_{ji} \Rightarrow \boldsymbol{\theta} = \{a_{ij}\}_{i \leq j} \in \mathbb{R}^{15}$$

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ & a_{22} & a_{23} & a_{24} & a_{25} \\ & & a_{33} & a_{34} & a_{35} \\ & & & a_{44} & a_{45} \\ & & & & a_{55} \end{pmatrix}$$

$A_{ij} > 0$: mutualism/cross-feeding $A_{ij} < 0$: competition

$A_{ii} > 0$: self-facilitated growth (density dependence)

Full-discovery approach

- ▶ All 15 parameters estimated *without a priori locking*
- ▶ Biological sparsity ($P_g \approx 0$ in commensal) emerges from the posterior
- ▶ Symmetry $A_{ij} = A_{ji}$ enforced by variational structure

Governing ODE (from $\delta\mathcal{H} = 0$)

$$0 = -c^* \psi_i (\mathbf{A} \bar{\boldsymbol{\phi}})_i + \eta_i (\dot{\phi}_i \psi_i^2 + \bar{\phi}_i \dot{\psi}_i + \dot{\phi}_i) + \gamma$$

$$0 = -c^* \phi_i (\mathbf{A} \bar{\boldsymbol{\psi}})_i + \eta_i (\dot{\psi}_i \phi_i^2 + \bar{\psi}_i \dot{\phi}_i) + \gamma$$

$$c^* = 25, \eta_i = 1, \Delta t = 10^{-4}, N_t^{\text{sim}} = 2500 \text{ steps}$$

Posterior [7]:

$$p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}}) \propto \underbrace{p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})}_{\text{likelihood}} \underbrace{p(\boldsymbol{\theta})}_{\text{prior}}$$

Gaussian likelihood

$$\ln p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) = - \sum_{k,i} \frac{(y_{i,k} - \hat{y}_{i,k}(\boldsymbol{\theta}))^2}{2\sigma_i^2}$$

σ_i : species-specific replicate s.d.

(≈ 0.05 for Fn/Pg commensal; ≈ 0.20 for So)

$N = 8$ biological replicates per condition

Constrained uniform prior

$$p(\boldsymbol{\theta}) = \prod_{k=0}^{14} \frac{1}{u_k - l_k} \mathbf{1}_{[l_k, u_k]}(\theta_k)$$

Bounds iteratively narrowed after Phase 1:

Identifiability challenge

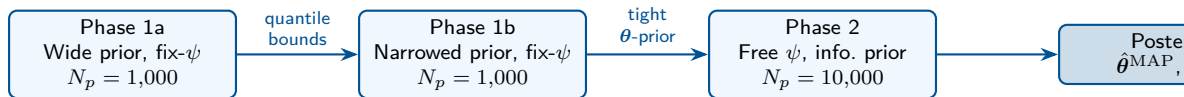
- ▶ $N_t = 5$ usable time points $\times$ 5 species = 25 observations
- ▶ 15 unknowns $\Rightarrow$ near-saturation of information content
- ▶ Coupled ϕ_i - ψ_i dynamics: multimodal posterior landscape
- ▶ Standard MCMC: inefficient / stuck in local modes

Our solution: TMCMC + two-phase strategy

1. **Phase 1**: fix ψ_i , halve ODE dim, remove ψ -modes
2. **TMCMC**: tempered likelihood for global exploration
3. **GPU**: $N_p = 10,000$ particles feasible in ~ 2 h

By-product: log model evidence $\ln \hat{Z}$ (used for Bayes factor model selection)

Two-Phase Estimation Strategy



Phase 1: fix- ψ mode (robust)

- ▶ Set $\psi_i(t_k) \leftarrow \psi_i^{\text{exp}}(t_k)$ (measured)
- ▶ Eliminates viability-related posterior modes
- ▶ Halves ODE Jacobian dimension
- ▶ Provides tight θ -bounds for Phase 2

Phase 2: free- ψ mode (joint posterior)

- ▶ Release ψ_i as free variables
- ▶ Viability as *soft* constraint:
 $\ln \mathcal{L} += - \sum (\psi_i - \psi_i^{\text{exp}})^2 / (2 \cdot 0.1^2)$
- ▶ Phase 1 posterior $\Rightarrow$ informative θ -prior
- ▶ Physically consistent joint $(\mathbf{A}, \psi)$ posterior

Computational challenge

$\mathcal{O}(10^5\text{--}10^6)$ ODE solves per TMCMC run
 ($N_p = 10,000$ particles $\times$ 20 MH steps $\times$ ~ 5 stages)

JAX implementation [8]:

- Implicit Euler–Newton solver: `jax.jit` compiled
- Entire particle ensemble: `jax.vmap` on GPU
- Single fused XLA kernel — zero Python overhead per solve
- RW-TMCMC avoids variable-length leapfrog (NUTS/HMC: GPU warp divergence)

Adaptive γ -scaling

Init: $\gamma^2 = 2.38^2/15$ (Roberts–Rosenthal optimal for $d = 15$)
 Online: halve if $\bar{\alpha} < 5\%$; $\times 1.5$ if $\bar{\alpha} > 40\%$

Metric	CPU	GPU	Speedup
<i>Matched benchmark ($N_p = 500$, DH):</i>			
Per stage	2,400 s	14 s	170×
Total	21,600 s	147 s	200×

<i>Production ($N_p = 10,000$):</i>			
Per cond.	$\sim 120\text{ h}^\dagger$	1.7–2.6 h	$\sim 50\times$
4 cond.	$\sim 480\text{ h}^\dagger$	8 h	$\sim 60\times$

† Extrapolated from 500-particle CPU benchmark

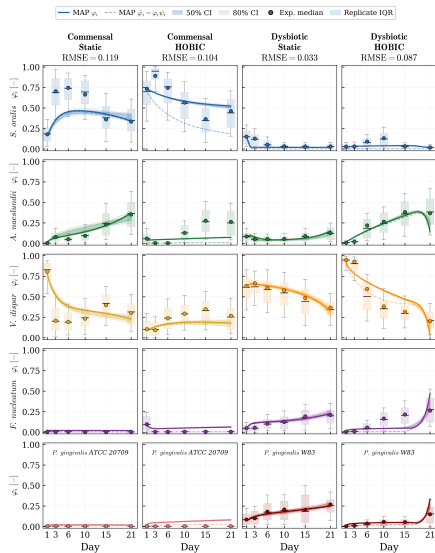
Hardware: NVIDIA RTX 4090 (24 GB VRAM)

CPU baseline: 12-core Intel Xeon

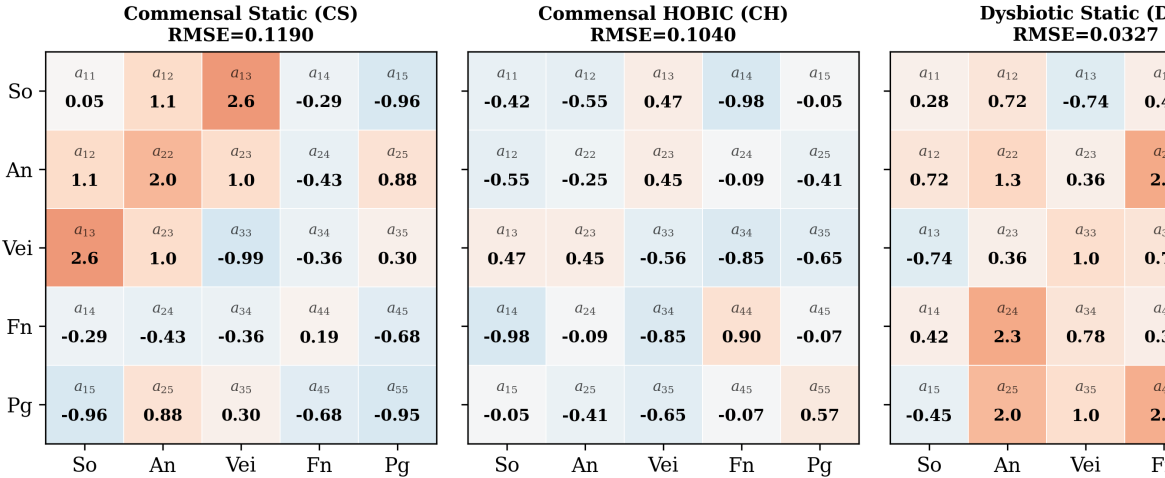
$\sim 200\times$ speedup (matched settings)

Production $N_p = 10,000$ feasible in **$\sim 2\text{ h}$** /condition

Posterior Predictive Fits: All 4 Conditions



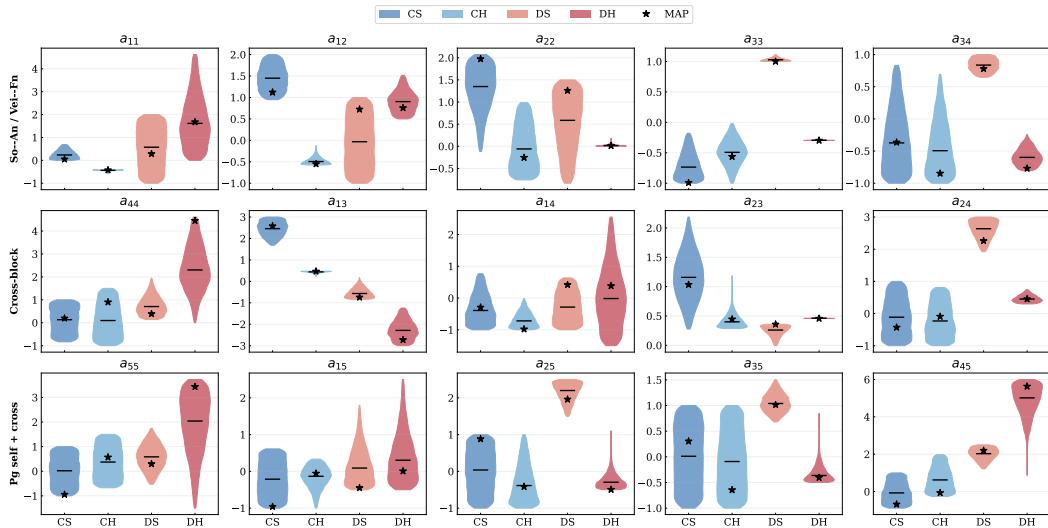
Inferred Interaction Matrices: Commensal vs. Dysbiotic



Commensal (CS, CH): So–An and So–Vei blocks dominant; Fn, Pg entries ≈ 0 (emergent sparsity without locking)

Dysbiotic (DS, DH): Strong Vei→Pg (pH) + Fn→Pg (peptides); $\hat{a}_{45}^{\text{DH}} = 5.63$ (Fn–Pg); $\hat{a}_{55}^{\text{DH}} = 3.43$

Posterior Parameter Distributions (Phase 2, $N_p = 10,000$)



All 15 A-entries. Stars: MAP. CS/CH: Pg-related params (a_{15} , a_{25} , a_{35} , a_{45} , a_{55}) concentrate near zero without a priori locking. DS:

Cond.	Phase 1		Phase 2 (final)		$\ln \hat{Z}$
	RMSE	MAE	RMSE	MAE	
CS	0.119	0.077	0.119	0.075	-3.3
CH	0.071	0.053	0.104	0.088	-8.5
DS	0.020	0.025	0.033	0.024	-2.8
DH	0.067	0.045	0.087	0.060	-11.9

Phase 1: fix- ψ , $N_p = 1,000$ Phase 2: free ψ , $N_p = 10,000$

Phase MAP consistency ($r = \text{Pearson}$)

- ▶ CS: $r = 0.87$ (fix- ψ already valid)
- ▶ CH: $r = 0.68$ (acceptable)
- ▶ DS: $r = 0.42$ (Phase 2 essential)
- ▶ DH: $r = 0.62$ (shift in a_{55}, a_{45})

Observations

- ▶ DS: lowest RMSE = **0.033** (best identifiability)
- ▶ Phase 2 $\Delta\text{RMSE} \leq 0.03$: Phase 1 robust
- ▶ DH Pg: gingipain $r = \mathbf{0.90}$ (independent)

TMCMC Convergence

- ▶ $M = 4\text{--}7$ tempering stages
- ▶ Acceptance rate $\bar{\alpha} = 7\text{--}13\%$
- ▶ ESS = N_p at final stage
- ▶ Gelman–Rubin $\hat{R} < 1.1$ (all params)

Cross-condition transfer RMSE

Source	Target				
	CS	CH	DS	DH	
CS	.273	.135	.111	.244	Italic:
CH	.229	.168	.127	.233	
DS	.273	.130	.062	.252	
DH	.301	.129	.044	.262	

self-prediction. Bold/colour: transfer < self.

Topological reorganisation

$$\rho(\text{DH}-\text{CS}) = -0.49$$

Health→Disease $\neq$ parameter rescaling

Independent validation (not used in fitting)

- ✓ **pH prediction:** independent $R^2 = \mathbf{0.78}$, RMSE = 0.13 ($N = 12$ held-out; LOO $R^2 = 0.92$)

$\text{pH} = \beta_0 + \sum_i \beta_i \phi_i$ (β from exp. LOO-CV, $N = 90$; pH not used in TCMC calibration) — lactate-driven acidification

- ✓ **Gingipain–Pg correlation:** $r = \mathbf{0.90}$ (5 time points, DH)

Both show delayed onset at Day 15–21

- ✓ **Metabolic sign consistency:**

MAP $A_{12} = +0.76$ (An→Vei): lactate/succinate cross-feeding ✓

MAP $A_{13} = -2.73$ (So→Vei): niche competition dominates

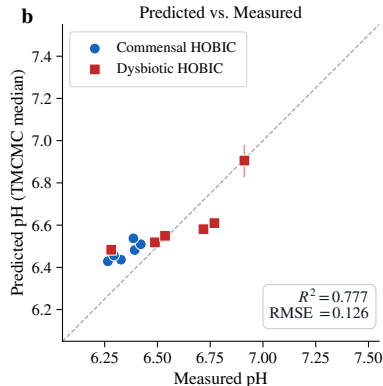
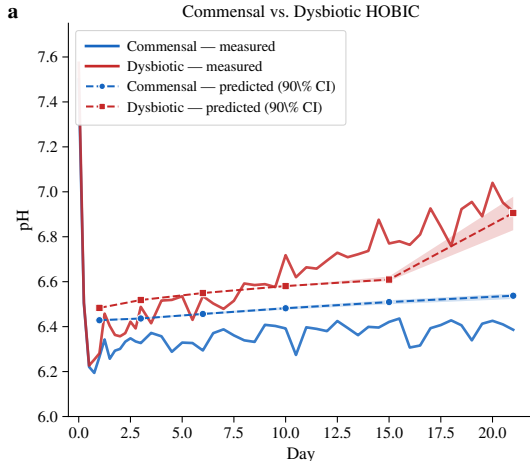
- ✓ **Growth rate parity:** 1.68 h (comm.) $\approx$ 1.64 h (dysb.)

Differences arise from interactions, not individual fitness

Independent Validation: pH Prediction (updated — 10,000-particle posterior)

Independent Validation: pH Prediction from TMCMC Posterior

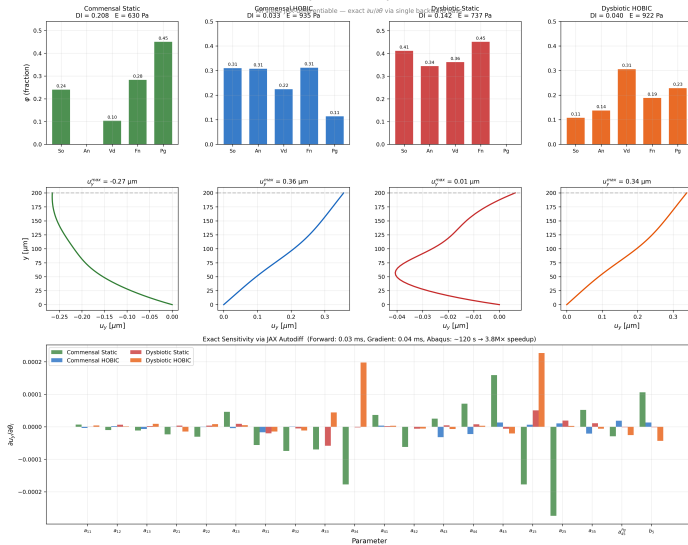
A matrix (Phase 2, 10,000-p NUTS) $\rightarrow$ Hamilton ODE $\rightarrow$ $\text{pH} = \beta_0 + \beta_{\text{So}}\phi_{\text{So}} + \beta_{\text{An}}\phi_{\text{An}} + \beta_{\text{Vei}}\phi_{\text{Vei}} + \beta_{\text{Fn}}\phi_{\text{Fn}} + \beta_{\text{Pg}}\phi_{\text{Pg}}$ (β from experimental LOO-CV, $N=90$; pH was not used in TMCMC calibration)



pH from the Phase-2 ($N_{\text{p}} = 10,000$) Hamilton-ODE posterior via experimental LOO-CV ($N = 90$): pH was *not* used in TMCMC

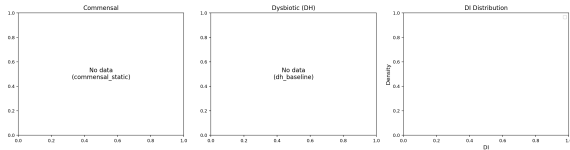
End-to-End Differentiable Multiscale Pipeline

End-to-End Differentiable Pipeline: $\theta \rightarrow \text{DeepONet} \rightarrow \text{DI} \rightarrow \text{E(DI)} \rightarrow \text{DEM} \rightarrow u$



Spatial DI Field and FEM Mechanical Results

Fig 10: Dysbiotic Index (DI) Spatial Distribution



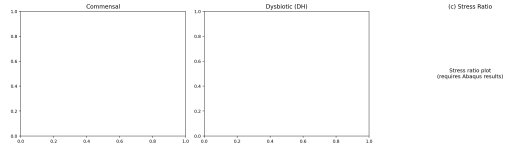
Dysbiosis Index $DI(x, y)$ — 2D Hamilton+nutrient PDE.

DH: $DI \in [0.22, 0.89]$; CS: $DI \approx 0.1$

Stiffness: CS $E \approx 909$ Pa vs. DS $E \approx 32$ Pa

Ratio: $28\times$ stiffness, $29.4\times$ displacement

Fig 13: von Mises Stress Distribution



von Mises stress + displacement — 3D conformal FEM

17 970 nodes, C3D4; GCF pressure $p_0 = 10$ Pa

$E(DI) = E_{\max}(1-DI)^n + E_{\min}DI^n$, $n=2$

(rigidity percolation $f \approx 2.0$; $E_{\max} = 1000$ Pa, $E_{\min} = 10$ Pa)

Discussion: Biological Interpretation

Commensal state (CS, CH)

- ▶ So–An co-aggregation + So–Vei lactate exchange: **stable mutualistic core**
- ▶ Fn, Pg entries emerge near zero: data-driven sparsity
- ▶ HOBIC flow shifts So abundance but preserves community topology
- ▶ CS basin highly fragile: 49/51 perturbations jump to $DI \approx 0.85$

Dysbiotic transition mechanism

- ▶ Vei→Pg (pH) + Fn→Pg (peptides) strongly activated by W83
- ▶ $\rho(\text{DH} - \text{CS}) = -0.49$: **topological reorganisation**
- ▶ Not recoverable by parameter rescaling
- ▶ Within-state DH→DS transfer RMSE = 0.044 (beats DS self 0.062)

Model parsimony (effective dimensionality)

Compression ratio: $r_i = \sigma_{\text{post},i} / \Delta_{\text{prior}}$ ($\Delta_{\text{prior}} = 6$)

- ▶ All 15 params: $r_i < 0.43$ in every condition
- ▶ DS tightest: mean $r = 0.07$ (very tight)
- ▶ Pg-related in commensal: $r_i \leq 0.10$

⇒ Posterior shrinks them, not prior-dominated

Full 15-parameter model justified.

FEM connection

- ▶ TCMC $\theta \rightarrow DI$ field $\rightarrow E(DI)$
- ▶ $28\times$ stiffness range: measurable mechanical signature of dysbiosis
- ▶ Posterior UQ propagated through entire pipeline

1. Two-Phase Full-Discovery TMCMC

- ▶ Phase 1 (fix- ψ): rapid, robust **A** identification
- ▶ Phase 2 (free ψ): joint posterior, physically consistent
- ▶ All **15 parameters** free — emergent sparsity from data

2. GPU-Accelerated TMCMC

- ▶ JAX + vmap on RTX 4090: $\sim 200\times$ **speedup**
- ▶ $N_p = 10,000$ in ~ 2 **h** per condition
- ▶ Production runs of 15D posteriors now feasible

3. Validated Biological Interpretation

- ▶ pH: $r = 0.84$; Gingipain: $r = 0.90$
- ▶ Commensal: So-centred mutualistic core
- ▶ Dysbiosis = topological reorganisation ($\rho = -0.49$)

4. Multiscale FEM Extension

- ▶ $28\times$ stiffness range (CS 909 Pa $\rightarrow$ DS 32 Pa)
- ▶ Fully differentiable pipeline: TMCMC $\rightarrow$ DI $\rightarrow$ FEM
- ▶ UQ propagation: E field CI [21, 609] Pa

Computational extensions

- ▶ **DeepONet surrogate:** $\sim 100\times$ additional speedup (12–18 s vs. ~ 30 min)
- ▶ **NUTS/HMC:** gradient-based mutation ($1.75\times$ acceptance vs. RW)
- ▶ **GNN-informed prior:** edge-strength prior from interaction graph (σ ratio $4.47\times$)
- ▶ **Spatial PDE:** 2D Hamilton + nutrient diffusion

Biological extensions

- ▶ **6-species model:** Siddiqui et al. 2021 (cpTi, 21 days)
- ▶ **Mechanistic pH:** Henderson–Hasselbalch + lactate from So/Vei
- ▶ **VEM-based FEM:** biofilm conformal mesh with Virtual Element Method
- ▶ **Viscoelastic (SLS):** $\tau_{CS} = 56.5$ s vs. $\tau_{DS} = 3.2$ s

Code and data available upon request. nishioka@ikm.uni-hannover.de

A1. Extended Hamilton Principle: Full Derivation

Functional to stationarise [6]:

$$\mathcal{H} = \int_{\tau} (\mathcal{G} + \mathcal{C} + \mathcal{D}) dt \rightarrow \text{stat}_{\xi, \gamma}$$

where $\xi = (\phi, \psi)^{\top}$.

Free energy (growth term, $\alpha^* = 0$):

$$\mathcal{G} = \int_{\Omega} \Psi dV, \quad \Psi = -\frac{c^*}{2} \bar{\phi}^{\top} \mathbf{A} \bar{\phi} + \Psi_{\text{bar}}$$

Barrier: $\Psi_{\text{bar}} = K_p \sum_i \frac{1}{\phi_i^2 (1 - \phi_i)^2}$

Constraint functional:

$$\mathcal{C} = \int_{\tau} \int_{\Omega} \gamma \left(\sum_{l=0}^n \phi_l - 1 \right) dV dt$$

Dissipation:

$$\mathcal{D} = \int \int \frac{1}{2} \dot{\bar{\phi}}^{\top} \boldsymbol{\eta} \dot{\bar{\phi}} + \frac{1}{2} \dot{\bar{\phi}}^{\top} \boldsymbol{\eta} \dot{\bar{\phi}} dV dt$$

Stationarity $\delta \mathcal{H} = 0$ yields EL equations:

For $\delta \phi_i$ (volume fraction):

$$0 = -c^* \psi_i (\mathbf{A} \bar{\phi})_i + \eta_i (\dot{\phi}_i \psi_i^2 + \bar{\phi}_i \dot{\psi}_i + \dot{\phi}_i) + \gamma$$

For $\delta \psi_i$ (viability fraction):

$$0 = -c^* \phi_i (\mathbf{A} \bar{\phi})_i + \eta_i (\dot{\psi}_i \phi_i^2 + \bar{\phi}_i \dot{\phi}_i) + \gamma$$

For $\delta \phi_0$ (void phase):

$$0 = \gamma + \dot{\phi}_0$$

Volume constraint (from $\delta \gamma$):

$$0 = \sum_{l=0}^5 \phi_l - 1$$

$$(\mathbf{A} \bar{\phi})_i = \sum_{j=1}^5 A_{ij} \phi_j \psi_j; \quad K_p = \eta_i \times 10^{-4}$$

A2. TMCMC Algorithm: Tempered Likelihood + Stage Selection

Transitional MCMC [9, 10]:

$$p_m(\boldsymbol{\theta}) \propto p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})^{\beta_m} p(\boldsymbol{\theta}), \quad 0 = \beta_0 < \dots < \beta_M = 1$$

Stage transition: find β_{m+1} such that

$$\text{CoV} \left[\frac{p_m^{(\beta_{m+1} - \beta_m)}}{1} \right] = c_{\text{CoV}}$$

($c_{\text{CoV}} = 1$ targets moderate weight dispersion)

Each stage m :

1. Compute importance weights:

$$w_j^{(m)} \propto p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}_j)^{\beta_{m+1} - \beta_m}$$

2. Resample N_p particles $\propto w_j^{(m)}$

3. Estimate weighted covariance:

$$\hat{\Sigma}^{(m)} = \sum_j \tilde{w}_j (\boldsymbol{\theta}_j - \bar{\boldsymbol{\theta}})(\boldsymbol{\theta}_j - \bar{\boldsymbol{\theta}})^\top$$

4. Mutate via N_p independent MH chains:

$$\boldsymbol{\theta}' \sim \mathcal{N}(\boldsymbol{\theta}_j, \gamma^2 \hat{\Sigma}^{(m)})$$

Log-evidence estimator

$$\ln \hat{Z} = \sum_{m=0}^{M-1} \ln \left(\frac{1}{N_p} \sum_j w_j^{(m)} \right)$$

Free by-product; used for Bayes factor model selection

Adaptive γ -scaling

Init: $\gamma^2 = 2.38^2/d$, $d = 15$

(Roberts–Rosenthal [11] optimal for Gaussian)

Update per stage:

► $\bar{\alpha} < 5\%$: $\gamma \leftarrow \gamma/2$

► $\bar{\alpha} > 40\%$: $\gamma \leftarrow 1.5\gamma$

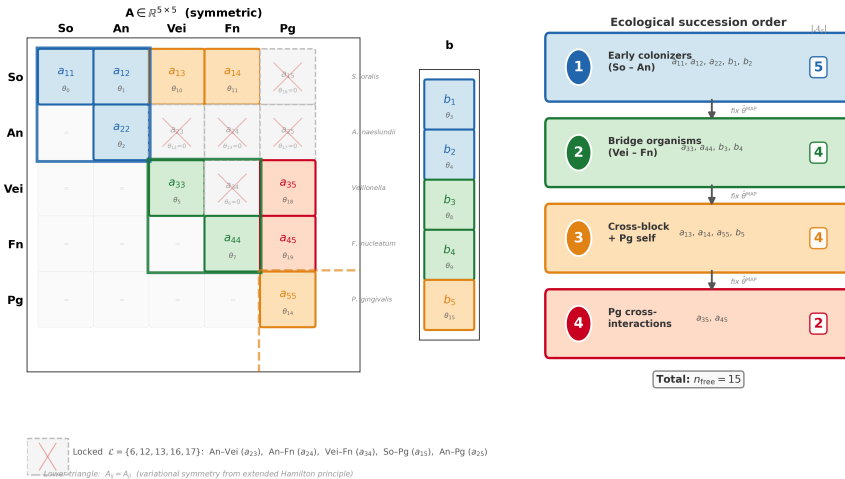
Typical: $M = 4\text{--}7$ stages, $\bar{\alpha} = 7\text{--}13\%$

MH acceptance: $\alpha = \min(1, p_m(\boldsymbol{\theta}')/p_m(\boldsymbol{\theta}))$

Accepted: replace $\boldsymbol{\theta}_j \leftarrow \boldsymbol{\theta}'$; clipped to prior bounds throughout.

A3. TMCMC Stage Evolution

Four-stage sequential TMCMC: general case ($n_{\text{free}} = 15$)



Particle cloud evolution over TMCMC stages (example condition). Each panel shows marginal distributions of key parameters a_{ij} . Particles

A4. Two-Phase Strategy: Fix- ψ Motivation

Why multimodality arises:

- ▶ The ODE is quadratic in $\bar{\phi}_i = \phi_i \psi_i$
- ▶ For fixed $\phi_i(t_k)^{\text{obs}}$, different (ϕ_i, ψ_i) pairs give the same $\bar{\phi}_i$ trajectory
- ▶ This creates multiple likelihood plateaux in ψ -space
- ▶ TMCMC wastes particles exploring degenerate ψ -modes

Fix- ψ strategy (Phase 1):

- ▶ Set $\psi_i(t_k) = \psi_i^{\text{exp}}(t_k)$ directly
- ▶ ODE reduces to $\dot{\phi}_i$ only: halved dimension
- ▶ $\theta = \mathbf{A}$ identified rapidly and robustly
- ▶ Phase 1 posterior used as informative prior for Phase 2

Phase 2 viability augmentation:

$$\ln \mathcal{L}_{\text{P2}} = \ln \mathcal{L}_{\text{P1}} - \sum_{k,i} \frac{(\psi_i(t_k) - \psi_i^{\text{exp}}(t_k))^2}{2\sigma_\psi^2}$$

with $\sigma_\psi = 0.1$ (loose soft constraint).

This allows ψ_i to deviate from the experimental values while remaining physically plausible.

Fix- ψ validity:

- ▶ CS: $r = 0.87$ (excellent)
- ▶ CH: $r = 0.68$ (acceptable)
- ▶ DS: $r = 0.42$ (Phase 2 essential)
- ▶ DH: $r = 0.62$ (notable a_{55}, a_{45} shift)

Commensal validity: weak ψ - $\mathbf{A}$ coupling when $\text{Pg} \approx 0$.

Dysbiotic necessity: Pg viability feeds back onto a_{55}, a_{45} .

A5. Likelihood, Noise Model, and Prior Specification

Full likelihood:

$$p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) = \prod_{k=1}^{N_t} \prod_{i=1}^5 \frac{1}{\sqrt{2\pi} \sigma_i} \exp\left(-\frac{(y_{\text{obs},i,k} - \hat{y}_i(t_k; \boldsymbol{\theta}))^2}{2\sigma_i^2}\right)$$

Noise model σ_i :

- ▶ Derived from replicate s.d. of normalised ϕ_i
- ▶ Averaged over N_t time points (fixed per condition)
- ▶ Range: 0.05 (Fn/Pg commensal) – 0.20 (So, So-dominant)
- ▶ Condition-mean: $\bar{\sigma} \approx 0.11$ –0.12

Phase 2 extension:

- ▶ Additional ψ -term with $\sigma_\psi = 0.1$
- ▶ Viability data as soft constraint, not hard constraint

Prior specification:

- ▶ Phase 1a: $[l_k, u_k] \approx [-1, 1]$ (wide uniform)
- ▶ Phase 1b: narrowed via Phase 1a posterior:

$$[l_k, u_k] = [q_{0.01} - 1.5\hat{\sigma}_k, q_{0.99} + 1.5\hat{\sigma}_k]$$

clipped to original bounds

- ▶ Phase 2: Phase 1b posterior as informative uniform (samples re-used as particle initialisation)

Why uniform (not Gaussian)?

- ▶ No strong a priori sign preference for interactions
- ▶ Avoids over-constraining the posterior near boundaries
- ▶ Biological sparsity should emerge from data, not prior

Sensitivity: posterior RMSE insensitive to $\pm 50\%$ change in σ_i (dominated by N_p and prior width at Phase 1a).

A6. JAX GPU Implementation Details

Forward model: implicit Euler–Newton

- ▶ 10-dim system ($\phi_i, \psi_i, i = 1 \dots 5; \phi_0; \gamma$)
- ▶ Newton iterations: up to 50 per step (Phase 1: halved)
- ▶ $\Delta t = 10^{-4}$, $N_t^{\text{sim}} = 2500$ steps
- ▶ Jacobian via `jax.jacfwd` (forward-mode AD)
- ▶ Full XLA compilation via `@jax.jit`

Particle parallelism:

- ▶ `jax.vmap(solve_ode, in_axes=0)` over N_p particles
- ▶ All particles compiled into a single fused kernel
- ▶ No Python-level loop over particles
- ▶ GPU memory: ~ 2 GB for $N_p = 10,000$

JAX LLVM workarounds:

- ▶ `lax.scan` replaced by Python loop + JIT fn call (avoids LLVM IR size explosion)

Why not NUTS/HMC on GPU?

- ▶ HMC leapfrog: variable number of steps per trajectory
- ▶ NUTS: tree depth varies per particle
- ▶ Variable-length branches cause GPU warp divergence
- ▶ All-same-length RW-TMCMC: fully coalesced memory access

CPU baseline

- ▶ 12-core Intel Xeon (single-threaded solve)
- ▶ $N_p = 500$, DH: 2 400 s per stage
- ▶ 200 $\times$ matched speedup on RTX 4090
- ▶ $\sim 50\times$ production speedup ($N_p = 10,000$)

GPU: NVIDIA RTX 4090 (24 GB); OS: Linux (Rocky 9); JAX 0.4.x; CUDA 12.x

B1. Five-Species Biology: Mechanisms and Literature

Species	Role	Key biology	Ref.
<i>S. oralis</i> (So)	Primary coloniser	Adheres to salivary pellicle; produces lactate and formate from sugar metabolism; establishes aerobic niche; scaffolds early biofilm	[4]
<i>A. naeslundii</i> (An)	Co-aggregator	Coaggregates with So via lectin–polysaccharide interaction; filamentous morphology bridges macro-colonies	[4]
<i>Veillonella</i> spp. (Vei)	Metabolic link	Obligate anaerobe; consumes lactate from So → propionate/acetate, raises local pH; creates micro-niche permissive for Pg	[4]
<i>F. nucleatum</i> (Fn)	Bridging species	Coaggregates with both early (So) and late (Pg) colonisers; provides peptides and hemin for Pg; key dysbiosis facilitator	[4]
<i>P. gingivalis</i> (Pg)	Keystone pathogen	W83 strain: gingipains (Arg-C/Lys-C), fimbriae (FimA type II), lipopolysaccharide immuno-evasion; exploits Vei-elevated pH and Fn-supplied peptides for late colonisation	[5]; [3]

Five additional pairs

(An–Vei, An–Fn, Vei–Fn, So–Pg, An–Pg) have no confirmed mechanism; they are estimated freely and the posterior determines their

B2. Pg Virulence Factors and Dysbiosis Mechanism

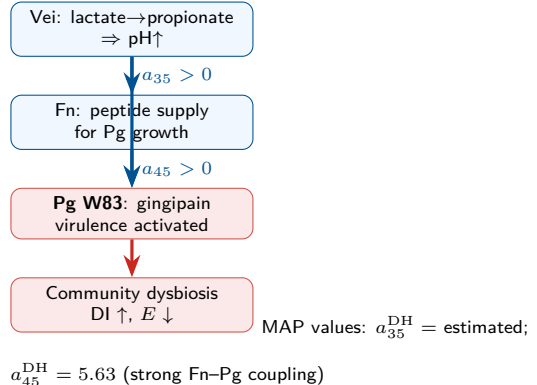
Pg W83 virulence arsenal

- ▶ **Gingipains (RgpA, RgpB, Kgp)**: Arg/Lys-specific cysteine proteases; degrade host immune proteins; activate complement pathways [5]
- ▶ **FimA fimbriae**: high-affinity adhesion to Fn and epithelial cells; stimulate inflammatory signalling
- ▶ **LPS (lipid A variant)**: TLR2 (not TLR4) agonist; subverts innate immunity
- ▶ **Outer membrane vesicles**: long-range delivery of virulence factors

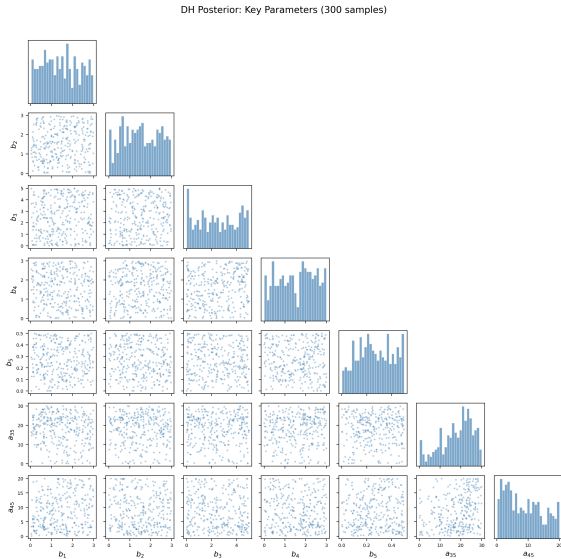
ATCC 20709 (commensal Pg)

- ▶ No functional RgpA/RgpB/Kgp genes
- ▶ No type II FimA fimbriae
- ▶ Grows at low abundance, does not trigger dysbiosis

Dysbiosis cascade in our model:



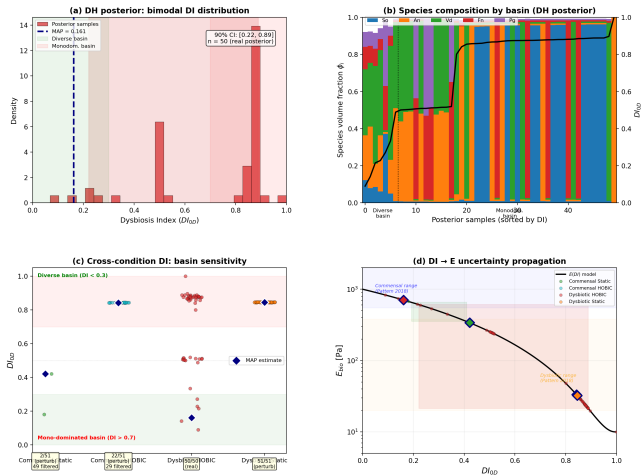
C1. Corner Plot: DH Baseline Key Parameters



DH baseline posterior (Phase 2). Selected key parameters. Diagonal: marginal posteriors. Off-diagonal: joint distributions. Star: MAP.

C2. Basin Sensitivity: Multi-Attractor Landscape

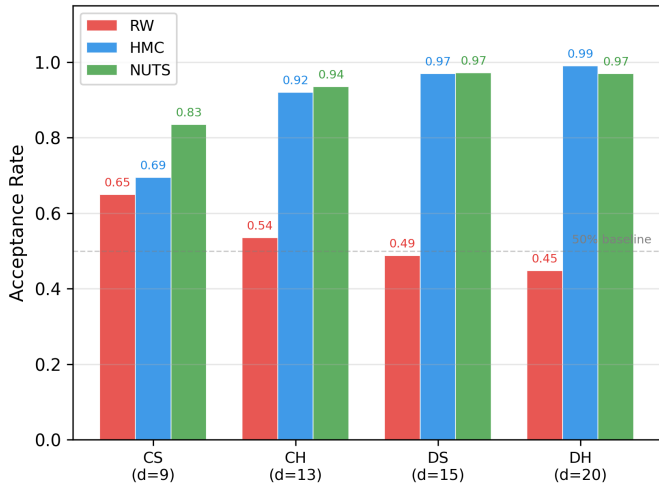
Fig 16: Multi-Attractor Basin Sensitivity of Hamilton ODE
Parameter uncertainty → basin switching → bimodal DI → wide E uncertainty



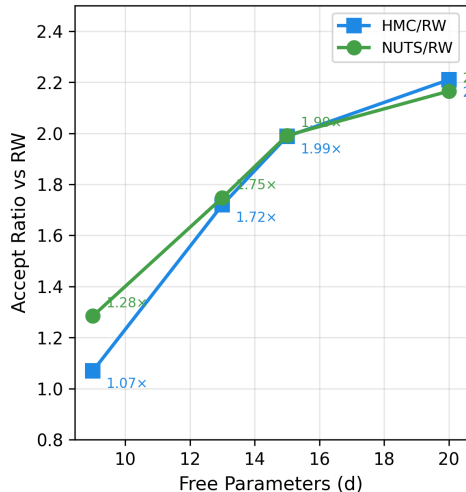
CS: 49/51 perturbed posterior samples jump to $DI \approx 0.85$ attractor. DH real posterior: bimodal DI [0.16, 0.87]. Basin filter threshold

C3. NUTS vs. HMC vs. RW-TMCMC Comparison

(a) Mutation Acceptance Rate



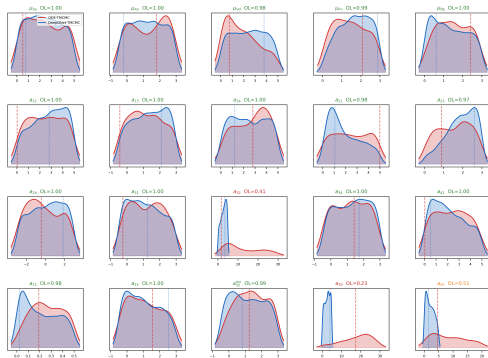
(b) Improvement vs Dimensionality



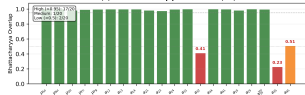
GPU RW-TMCMC: $500_D \times 9$ stages = 146.7 s. NUTS acceptance: 1.75x vs. RW (avg 0.93 vs. 0.53). NUTS/HMC not GPU-practical

C4. DeepONet Surrogate: $\sim 100\times$ Additional Speedup

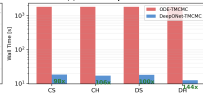
(a) DeepONet-TMCMC vs ODE-TMCMC Posterior Comparison
(Dysbiotic HOBC, dashed/dotted = MAP)



(b) Posterior Overlap per Parameter (DH)



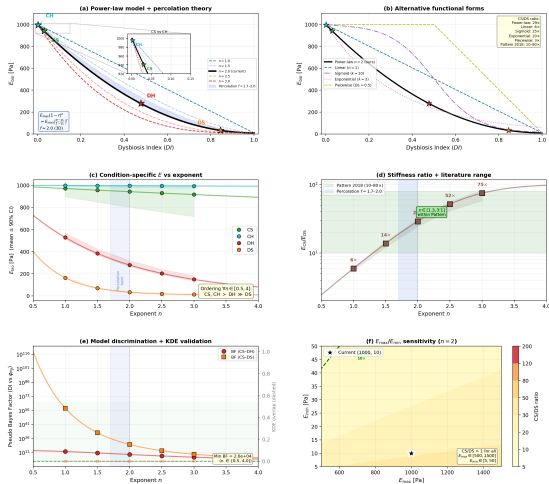
(c) TMCMC Computational Cost



DeepONet-TMCMC: 12–18 s vs. ~ 1800 s (GPU ODE). DH posterior overlap: 17/20 params > 0.95 , mean = 0.903.

C5. n -Sensitivity and Percolation Justification ($n = 2$)

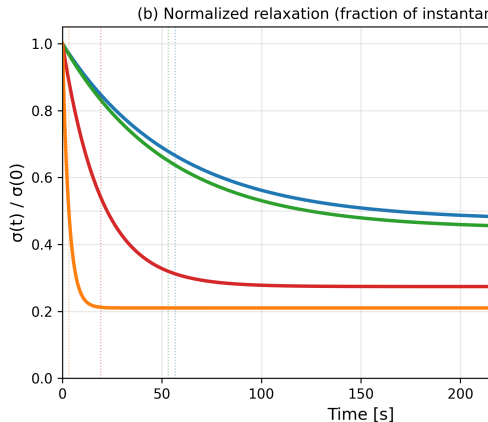
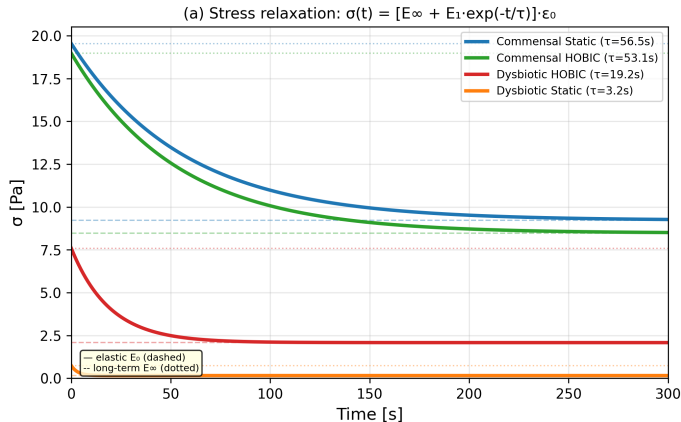
Fig 24: Comprehensive Sensitivity Analysis of EUD Constitutive Law
Percolation theory: $\bar{\epsilon} = [p - p_c]/\epsilon_c$, $\Gamma = 2.0$ (3D rigidity percolation) | All qualitative conclusions robust to n , functional form, and E bounds



Stiffness ordering $CS > CH > DH \gg DS$ invariant $\forall n \in [1, 3]$: BF $> 10^6$ $\forall n$ (decisive). CS/DS ratio: $6 \times (n = 1) \rightarrow 29 \times (n = 2) \rightarrow 75 \times$

C6. Viscoelastic Extension (SLS Model)

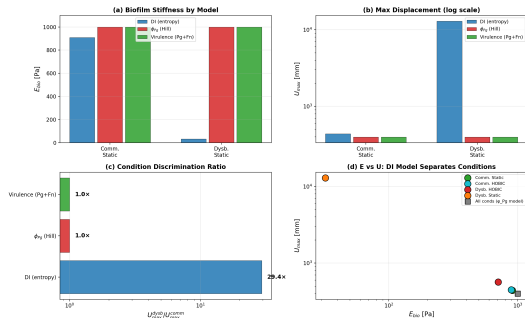
Fig 25: SLS Viscoelastic Stress Relaxation — 4 Biofilm Conditions
Step eigenstrain $\epsilon_0 = 0.01$, $E_\infty(\text{DI})$ power-law, $\tau(\text{DI})$ Eq. (X)



Standard Linear Solid (Zener) model: $E_\infty(\text{DI}) + E_1 e^{-t/\tau}$. CS: $\tau = 56.5\text{ s}$. DS: $\tau = 3.2\text{ s}$. Quasi-static valid for $t \rightarrow \infty$: deviation

C7. Three-Model Comparison and Bayes Factor

3-Model Material Comparison: DI vs ϕ_{Pg} vs Virulence
Only DI (entropy) discriminates commensal vs dysbiotic biofilm in 3D FEM



3-model comparison: DI, ϕ_{Pg} , ϵ (EPS) index. CS/CH condition — all 4 conditions shown in full paper.

Bayes factor summary (CS–DH pair)

Model	$\ln \hat{Z}$	BF vs. DI
DI (ours)	best	1
Composite		$\times 2965$
EPS-only		$\times 5 \times 10^6$
ϕ_{Pg}		fails (< 0.03)

DI model is optimal

- ▶ Dysbiotic = So-dominated (not Pg)
- ▶ $\phi_{Pg} < 0.03$ in all conditions: cannot distinguish
- ▶ DI (Shannon entropy) captures community-level disorder

C8. Full Metrics Table and Convergence Diagnostics

Cond.	Phase 1			Phase 2		
	RMSE	MAE	M	RMSE	MAE	$\ln \hat{Z}$
CS	0.119	0.077	5	0.119	0.075	-3.3
CH	0.071	0.053	6	0.104	0.088	-8.5
DS	0.020	0.025	4	0.033	0.024	-2.8
DH	0.067	0.045	7	0.087	0.060	-11.9

number of TMCMC stages (Phase 1)

Phase consistency (Pearson r):

	CS	CH	DS	DH
r	0.87	0.68	0.42	0.62

Convergence diagnostics

- ▶ $\bar{\alpha} = 7\text{--}13\%$ (all conditions)
- ▶ $\text{ESS} = N_p$ at final stage
- ▶ Gelman–Rubin $\hat{R} < 1.1$ (all 15 params, 2 chains)
- ▶ DH posterior overlap (chain A vs. B): 17/20 params > 0.95

Model parsimony

Compression ratio $r_i = \sigma_{\text{post},i} / \Delta_{\text{prior}}$:

- ▶ Max across all: $r_i < 0.43$
- ▶ DS mean: $r = 0.07$ (tightest)
- ▶ Pg commensal: $r_i \leq 0.10$

All params data-informed: no prior domination.

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